Enumerative Combinatorics — problem 5.49

A. Engel — Problem-Solving Strategies · 1998

Along a one-way street there are n parking lots. One-by-one n cars numbered 1 to n enter the street. Each driver i heads to his favorite parking lot a_i, and, if it is free, he occupies it. Otherwise, he continues to the next free lot and occupies it. But if all succeeding lots are occupied, he leaves for good. How many sequences a_i are such that every driver can park (M.D. Haiman, J. Algebraic Combinatorics, 3, 17–76 (1994) and SPMO 1996) ?